Communication Tone, Pull Request Outcomes, and Contributor Return in Open Source
Part 3: Final Results

Dimitrios Stamatios Bouras
2501112125

Quantitative Analysis of Open Source Software

Spring 2026

The Question

Core question (observational)

Do communication signals in pull request discussion associate with whether a PR gets merged and whether the author comes back?

- RQ1 Which signals associate with **PR merge outcome**?
- RQ2 Which signals associate with **author return** at 30 / 90 / 180 days?
- RQ3 Are **newcomers** affected differently from experienced contributors?
- RQ4 Does **politeness** buffer the effect of isolated harsh language?

Signals measured per PR

Toxicity (max), sentiment, anger — transformer models | Politeness — lexical heuristic

The Dataset

Scale

- **12,098** focal PRs (Jan 2024 – Mar 2025)
- **3** repositories: Kubernetes, Next.js, scikit-learn
- **2,210** PR authors; **100k+** scored comments

Key rates (mean $\pm$ SD)

- Merge: **75.7%** Return 30d: **76.8%**
- Return 90d: **81.3%** 180d: **83.5%**
- Prior PR count: mean 65.7, SD 96.1

Key descriptive fact

Only **1.0%** of PRs have any comment with toxicity ≥ 0.05. These are well-moderated projects.

Data leakage prevention

A **6-month lookback** (Jul–Dec 2023) precedes the focal window. This ensures contributor history is available before any focal PR is observed — preventing leakage into newcomer classification.

Methodology

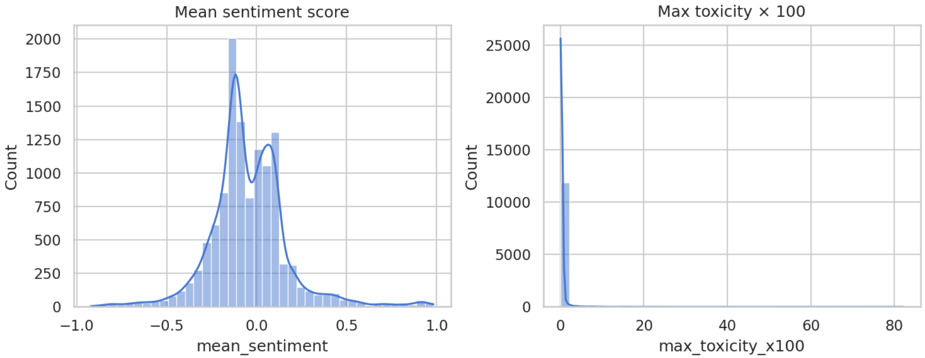
Variable types and model choice

- Outcomes (*merged*, *returned*) are **binary** $\rightarrow$ logistic regression
- NLP signals are **continuous** $\in [0, 1]$ or $[-1, 1]$
- Controls: prior PR count, comment count, participants, response delay, repo fixed effects
- HC3 robust standard errors; VIF 1.01–1.25 (no multicollinearity)

Key cleaning decisions

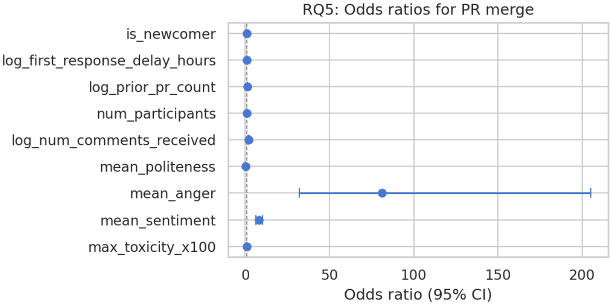
- Bots filtered by login pattern
- Comments $<$ minimum length discarded
- Code blocks and URLs stripped before NLP scoring
- Only **received** comments scored (excludes PR author's own text)
- Duplicate comment texts deduplicated before model inference

What the Signals Look Like



- **Sentiment** is well-distributed across PRs — a usable continuous signal
- **Max toxicity** is extremely right-skewed: almost all PRs sit near zero
- This is why averaging toxicity fails and why max toxicity and binary flags are needed

RQ1: What Predicts Merge?



$N = 12{,}098$, McFadden pseudo-$R^2 = 0.118$

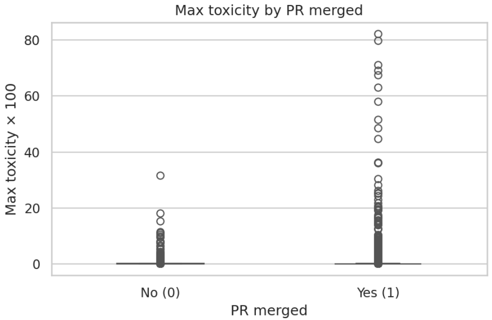
Significant signals

- **Sentiment** strongly positive
  coef $= +2.10$, $p < 10^{-60}$
- **Anger** positive
  coef $= +4.40$, $p < 10^{-20}$
- **Politeness** negative
  coef $= -1.42$, $p < 10^{-12}$

Note

Anger and politeness direction explained on next slide.

Why the Anger and Politeness Findings Make Sense



Merged PRs carry the high-toxicity outliers — raw data confirms the direction.

Anger → more merging

"This is completely broken" scores as *anger* in a social-media model. In code review it signals **engaged review** pushing toward a decision.

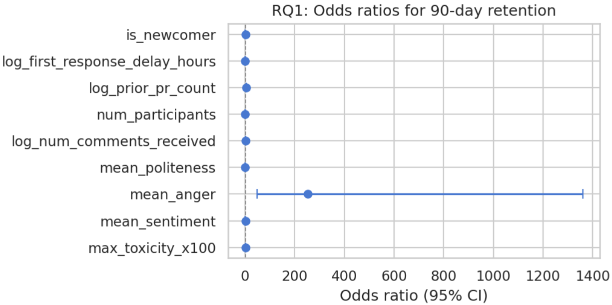
Politeness → fewer merges

"Maybe consider...", "could you perhaps..." marks **soft rejection**, not approval.

Key lesson

NLP signals carry real information, but not what they were designed to capture. Domain context is essential.

RQ2: What Predicts Return?



$N = 12{,}098$, McFadden pseudo-$R^2 = 0.498$

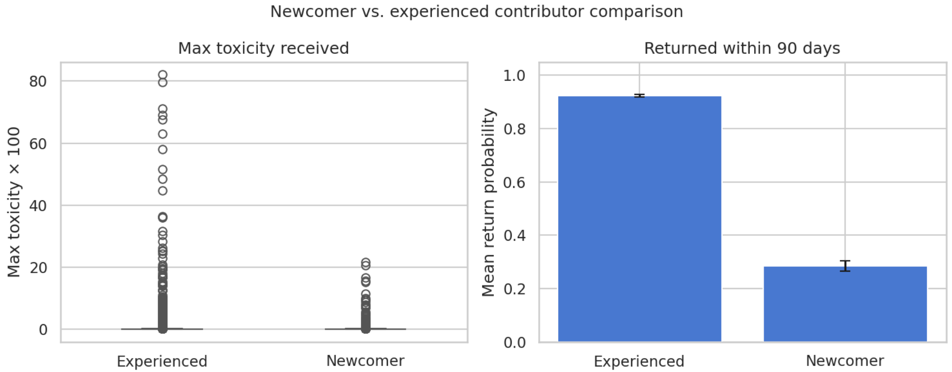
Significant signals

- **Prior PR count** dominates
  coef $= +1.51$, $p < 10^{-179}$
- **Anger** still positive
  coef $= +5.54$, $p < 10^{-10}$
- Toxicity, sentiment, politeness: **not significant**

Main finding

Return is about **who you already are**, not the tone of one PR.

RQ3: Newcomers vs. Experienced Contributors



- **Left**: toxicity exposure is similar for both groups — newcomers are not treated more harshly
- **Right**: experienced contributors return at ∼92%; newcomers at ∼28%
- Newcomer × toxicity interaction: **not significant** ($p = 0.57$) — the gap is about

RQ4: Politeness as a Buffer

Does a polite discussion soften isolated harsh comments?

Interaction term: `max_toxicity` $\times$ `mean_politeness` in the 90-day return model.

- Interaction is **significant**: $p = 0.003$, coef $= +0.81$
- When overall discussion is polite, isolated harsh comments associate *less* negatively with return
- The overall culture of the discussion absorbs individual sharp moments

Interpretation

Context matters: one sharp comment in an otherwise civil discussion lands differently than the same comment in a hostile thread. Community tone is a cushion.

What We Learned

From the results

- Communication tone matters more for **merge** than for **return**
- Return is dominated by **contributor experience** — one bad discussion does not override years of history
- Overt toxicity is too rare in large moderated repos to drive retention outcomes on its own
- The politeness buffer suggests that **overall community culture** absorbs individual harsh moments

From the method

NLP signals trained on social media do not map cleanly to code review. Technical assertiveness scores as anger; tentative rejection scores as politeness. Domain context is essential for interpretation.

Broader takeaway

The biggest risk to losing contributors is not any single PR discussion — it is failing to onboard them deeply enough to begin with.

Limitations and Robustness

Design limitations

- Observational: associations only, not causes
- Return = opening another PR, not all activity
- Only 3 large English-language repositories
- NLP models trained on social media, not code review
- Author fixed effects failed (singular matrix)

Robustness checks

- Binary toxicity ≥ 0.05: retention OR significant ($p < 0.001$)
- Binary toxicity ≥ 0.10: not significant ($p = 0.15$)
- Results change with threshold — toxicity is sparse and operationalization-sensitive

On toxicity

The data do **not** support a simple "toxic comments drive people away" story.

Experience & Reflections

What I learned

- Running a full empirical pipeline end-to-end: API collection, NLP scoring, regression analysis
- Why data leakage matters — the lookback window was a non-obvious design decision that changed the results
- Off-the-shelf NLP models carry domain assumptions that can **invert** your expected findings

What was hardest

- Toxicity is so rare (1% of PRs) that standard analysis almost cannot find it
- Operationalization choices — mean vs. max, which threshold — changed significance
- Deciding what counts as a *result* vs. an artifact of the model

If I did it again

I would validate the NLP signals against domain-specific ground truth before trusting their direction.

Summary

Question	Finding	Strength
RQ1: signals and merge	sentiment +, anger +, politeness −	strong
RQ2: signals and return	prior experience dominates	very strong
RQ3: newcomer sensitivity	no differential toxicity effect	null result
RQ4: politeness moderation	buffers isolated harsh comments	moderate

One-sentence conclusion

Communication tone shapes **whether your PR gets accepted**, but **who you already are** determines whether you come back.

Questions?

Repository